Size Confounds Semantics: An Audit of SING's Null-Space Leakage Score

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Abstract

SING [10] ranks classifiers by how much class-relevant semantics leak into the linear head's null space. Their reported standout is a "DinoViT" pretrained with DINO, which they find leaks least. We checked that claim against public checkpoints and against size controls on the score itself. Among public ViT-L/16 models that match SING's code or wrappers, every candidate is supervised or SWAG-supervised; none reproduces the reported 84.0% top-1, and none is DINO. Holding the architecture fixed, the absolute-shift score |AS| grows by about a factor of 10 as the removed null subspace expands from d=24 to d=1048 on ResNet50, ResNeXt101 and DINOv2-L. Matching the models at d=24 already disagrees with SING's published order (Kendall τ=0.29). A pooled log-linear fit of |AS| against removed feature energy ρ gives an exponent of 0.43, close to the $\sqrt{\rho}$ geometric prediction, though separating energy from dimension remains incomplete. We therefore also report a size-fair score $\mathrm{AS}_{\mathrm{fair}}$, which divides null removal by an energy-matched random-subspace baseline. On all 13 SING models, $\mathrm{AS}_{\mathrm{SING}}$ recovers the published Fig. 6b ordering (τ=0.82), while $\mathrm{AS}_{\mathrm{fair}}$ does not (τ=−0.18), and two class-level claims reverse under bootstrap CIs. Planted-leakage recovery is weak for both scores. When null-space scores are used to compare architectures, we recommend reporting ρ and $\mathrm{AS}_{\mathrm{fair}}$ alongside raw AS.

1 Introduction

A linear classifier on features $f \in \mathbb{R}^m$ has a null space of dimension $m-c$ (for c classes) that leaves the logits unchanged. SING [10] treats class-relevant content in that null space as "leakage": material the head could have used, but did not. Absolute Shift (AS) measures how far a CLIP-aligned [5] translation of f moves once the null component is stripped out. SING ranks 13 ImageNet models by AS (and by the IS/AS ratio) and singles out a ViT labeled "DinoViT" as the least leaky [1].

Related null-space tools appear in OOD detection [2, 9] and overfitting scores [6]. Across architectures the null dimension $d=m-\mathrm{rank}(W)$ and the null energy fraction $\rho = \mathbb{E}\|V_n V_n^\top f\|^2/\mathbb{E}\|f\|^2$ both swing by orders of magnitude. If AS largely tracks how much energy is removed, then cross-model "semantics" rankings are partly rankings of null size.

What we find. SING's "DinoViT" is not identifiable as DINO: every matching public candidate is supervised, and none reproduces 84.0% top-1. Within-model AS(d) curves rise by $\approx$10$\times$ from d=24 to d=1048 on ResNet50, ResNeXt101 and DINOv2-L; BiT [3] is the exception, explained by translator cancellation. Energy and dimension are hard to separate. A pooled fit yields exponent a=0.43 on ρ against a $\sqrt{\rho}$ prediction of 0.5, yet fixed-ρ flatness is not clean. On 13 models, size-fair $\mathrm{AS}_{\mathrm{fair}}$ breaks SING's published ranking (τ: 0.82 → −0.18) and reverses several claims, with the caveats below. We treat $\mathrm{AS}_{\mathrm{fair}}$ as a reporting correction rather than a new leakage oracle.

2 Background and the DinoViT identity

SING AS in brief. Let $T(f) = Af + b$ be SING's released linear translator into CLIP space and V_n an orthonormal basis of $\ker(W)$ (tolerance 10^{-5}). For a text direction t, SING's per-image absolute shift is the change in angle $\theta(T(f), t)$ after removing $V_n V_n^\top f$. $\mathrm{AS}_{\mathrm{SING}}$ denotes the image-averaged |AS| under full null removal (class text unless noted).

Exp 0b: "DinoViT" is not DINO. SING reports DinoViT at 84.0% ImageNet top-1 (Supp. Table 2). The released models.yaml loads a timm AugReg ViT-L/16; the supplement's wrapper uses torchvision ViT attribute names. We ran five public ViT-L/16 candidates on the full 50k validation set (native and force-224): torchvision IMAGENET1K_V1 (79.67%), timm AugReg (85.85%), SWAG linear [7] (85.14%), SWAG E2E (88.07%), and HF google/vit-large-patch16-224 (81.98%). None lands in 84.0 ± 0.3 (Fig. 1). All five are supervised or SWAG weakly supervised. DINO never shipped a ViT-L with a 1000-way head [1]. So the DINO attribution in the headline result has no identifiable public checkpoint behind it.

Exp 0 (re-analysis). Digitizing SING Supp. Fig. 6b IS/AS bars and correlating against measured null dimension gives Spearman −0.46 (p=0.12) over 13 models (DenseNet null dim = 27 at tol 10^{-5}). Null energy is far more predictive: $\mathrm{Spearman}(\rho_{\mathrm{full}}, \mathrm{IS/AS}) = -0.94$ (p=1.9×10^{-6}). DeiT3-L [8] has null dim 24 yet the worst

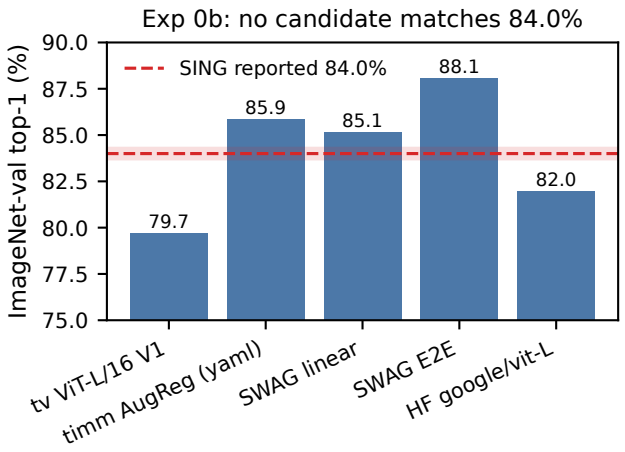


Figure 1: Exp 0b top-1 on ImageNet-val. No candidate matches SING's reported 84.0%.

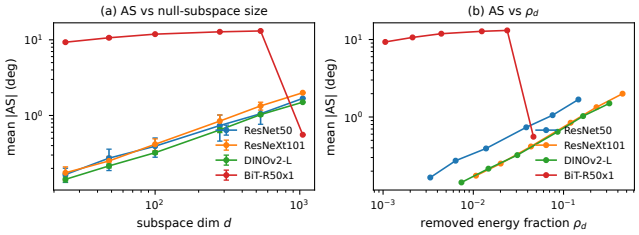


Figure 2: Exp 1: mean |AS| vs. subspace size d (a) and removed energy ρ_d (b). BiT collapses at the full null (translator cancellation).

ratio, so dimension alone cannot explain the ranking. That is why we move to a within-model test.

3 Within-model size dependence (Exp 1)

We fix a 10k ImageNet-val subset (10 images/class) and remove random d-dimensional subspaces of each model's own null space ($d \in \{24, 48, 100, 280, 536, 1048\}$, 20 seeds for $d<d_{\text{null}}$) for ResNet50, ResNeXt101, BiT-R50x1 and DINOv2-L (all d_{null}=1048), using SING's released translators.

Fig. 2 and Table 1 show r=AS(1048)/AS(24) $\approx$ 10 on 3 of 4 models. Matching d=24 over 8 models yields Kendall τ=0.29 (p=0.40) vs. the Fig. 6b order for |AS|. Both pre-registered GO conditions hold.

BiT anomaly. BiT's full-null AS collapses (r=0.06) even though partial subspaces give large AS (∼9–13°). The singular values of AV_n are extreme (top 284 vs. 5.6 for ResNet50): orthogonal null blocks nearly cancel through the translator when summed, so full-null AS understates BiT's per-direction sensitivity. The arithmetic is real; the collapse comes from translator conditioning.

Model	r	2σ band	R^2 vs d	R^2 vs ρ_d
ResNet50	10.15	[5.83, 14.48]	0.983	0.983
ResNeXt101	11.44	[6.93, 15.95]	0.977	0.977
DINOv2-L	10.48	[8.85, 12.10]	0.974	0.974
BiT-R50x1	0.060	[0.057, 0.063]	0.512	0.512

Table 1: Exp 1: r=AS(1048)/AS(24). Energy and dimension fits are collinear for random subspaces.

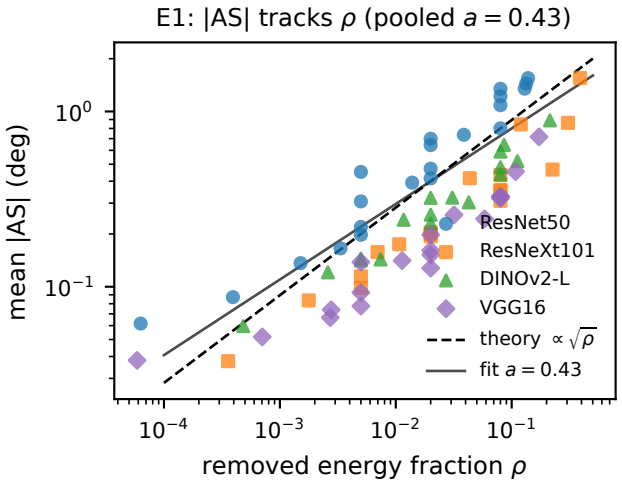


Figure 3: E1 cells: |AS| vs. ρ. Dashed: $\sqrt{\rho}$ theory; solid: fitted a=0.43.

A note on collinearity. For random subspaces ρ_d is nearly linear in d, so R^2 vs. d and vs. ρ_d coincide ($\approx$ 0.98). Exp 1 alone cannot separate energy from dimension.

4 Energy vs. dimension (E1)

Geometry. Suppose a linear translator maps a removed feature component of relative energy ρ into CLIP space. To first order the angular displacement scales with the removed magnitude $\|\Pi f\| \propto \sqrt{\rho}$ when the residual is small relative to $\|T(f)\|$. That predicts $|\text{AS}| \propto \rho^{1/2}$ at fixed geometry, i.e. an exponent of 0.5.

Design. On ResNet50, ResNeXt101, DINOv2-L, VGG16 and EVA02-L we (a) compare high-energy vs. low-energy null PCs at fixed $d \in \{24, 100, 280\}$ and (b) hit fixed $\rho \in \{0.5\%, 2\%, 8\%\}$ at several d by mixing top/bottom PCs. Pooled fit over 76 feasible cells:

$$\log|\text{AS}| \;=\; a\log\rho \;+\; b\log d \;+\; c_{\text{model}},$$

gives a=0.431 [0.430, 0.433], b=0.010 [0.008, 0.012], R^2=0.93 (Fig. 3).

High/low energy ratios at fixed d exceed 3× on all 4 evaluable models (e.g. ResNet 11–22×, ResNeXt $\approx$ 10×, DINOv2 3.7–5.1×, VGG16 6–10×). Fixed-ρ max/min across d still exceeds 1.5 on several cells (ResNet up to 2.29 at ρ=0.5%). EVA02-L (d_{null}=24) is not evaluable beyond its full-null cell. Energy dominates the fit and comes within $\approx$ 0.07 of the $\sqrt{\rho}$ exponent from the

geometry argument above. At the same time, AS still varies with d at fixed ρ, so the pre-registered ENERGY-DRIVEN flatness rule is only partly met. We treat the E1 outcome as mixed.

5 A size-fair score

Take an energy-matched random baseline: draw a Haar subspace of the full feature space that removes exactly the same energy $\mathbb{E}\|\Pi f\|^2$ as the null projector, and let $\mathrm{AS}_{\mathrm{rand}}$ be its mean $|\mathrm{AS}|$ (20 draws). Then

$$\mathrm{AS}_{\mathrm{fair}} \;=\; \frac{\mathrm{AS}_{\mathrm{null}}}{\mathrm{AS}_{\mathrm{rand}}}.$$

Values near 1 mean the null shifts about as much as a random equal-energy subspace; values $\gg 1$ mean the leakage is concentrated in the null. This is the E3-registered definition, reused at $\alpha{=}0$ for the E2 audit.

6 Planted leakage (E3)

We plant a CLIP concept direction (red/striped/water) into each of 8 models' null spaces at log-uniform $\alpha \in [0.01, 1]$ over 50 trials, leaving logits unchanged (max $|\Delta\text{logit}| \le 2.3{\times}10^{-5}$). Kendall τ vs. the planted-α ranking, concept-averaged: $\mathrm{AS}_{\mathrm{SING}}$ 0.386 $[0.324, 0.447]$; $\mathrm{AS}_{\mathrm{fair}}$ 0.071 $[0.004, 0.138]$. Neither recovers planted strength well; $\mathrm{AS}_{\mathrm{fair}}$ sits near chance. Per-concept means look similar (red 0.386/0.070, striped 0.403/0.061, water 0.369/0.083). A few quirks: DeiT3-L signed AS is negative at every α and $|\mathrm{AS}|$ shrinks as α grows; BiT $\mathrm{AS}_{\mathrm{SING}}$ reaches 17–20° at $\alpha{=}1$ while its $\mathrm{AS}_{\mathrm{fair}}$ at $\alpha{=}0$ is only 0.045–0.063; $\mathrm{AS}_{\mathrm{fair}}$ is non-monotonic in α for EVA02-L and ConvNeXt-B. Correcting for size and recovering a planted ranking are different goals, and the two scores trade off between them. We propose $\mathrm{AS}_{\mathrm{fair}}$ only as a size correction for cross-model AS comparisons. It is not meant as a calibrated leakage oracle.

7 Audit of SING's 13-model claims (E2)

On the same 10k subset plus a 200-image claim-class set (all 50 val images of jellyfish, porcupine, Arabian camel, sports car), we recompute $\mathrm{AS}_{\mathrm{SING}}$ and $\mathrm{AS}_{\mathrm{fair}}$ for all 13 released translators.

Ranking. Against the digitized Fig. 6b IS/AS order: $\tau(\mathrm{AS}_{\mathrm{SING}})$=0.821 ($p{=}1.3{\times}10^{-5}$); $\tau(\mathrm{AS}_{\mathrm{fair}})$=−0.179 ($p$=0.435). Dropping DenseNet: $\tau(\mathrm{AS}_{\mathrm{fair}})$=−0.364. Mean over 173 SING concepts: τ=0.641 vs. −0.026. $\mathrm{AS}_{\mathrm{SING}}$ recovers SING; the disagreement shows up only after the size correction (Figs. 4–5).

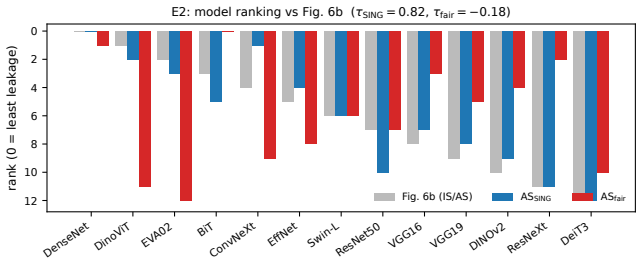


Figure 4: Ranks (0=least leakage). $\mathrm{AS}_{\mathrm{SING}}$ tracks Fig. 6b; $\mathrm{AS}_{\mathrm{fair}}$ does not.

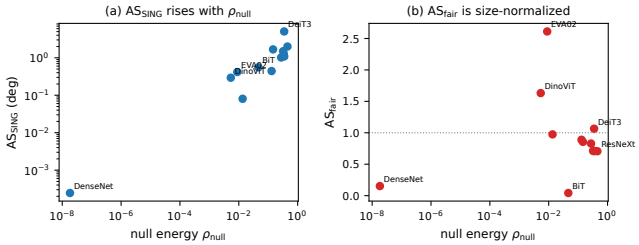


Figure 5: (a) $\mathrm{AS}_{\mathrm{SING}}$ rises with ρ_{null}. (b) $\mathrm{AS}_{\mathrm{fair}}$ removes that trend.

Claim reversals. Under $\mathrm{AS}_{\mathrm{fair}}$, claims C1–C7, C9, C11, C12 reverse relative to $\mathrm{AS}_{\mathrm{SING}}$ (DinoViT is no longer least leaky; ResNeXt is no longer among the worst). Two claims reverse under both scores, with bootstrap CIs:

- C8 (ResNet porcupine / sports-car ≥90th percentile): porcupine percentile 84.8 $[80.1, 87.7]$ ($\mathrm{AS}_{\mathrm{SING}}$), 68.4 $[64.5, 71.5]$ ($\mathrm{AS}_{\mathrm{fair}}$); sports car 61.0 $[56.1, 66.2]$ and 67.2 $[63.6, 71.4]$.
- C10 (DinoViT jellyfish $\gg$ camel): jellyfish−camel CI $[-0.205, -0.010]$ ($\mathrm{AS}_{\mathrm{SING}}$) and $[-1.403, -0.458]$ ($\mathrm{AS}_{\mathrm{fair}}$)—opposite in sign to SING's claim.

Caveats. (i) C1/C2 are IS/AS-ratio claims tested here on $|\mathrm{AS}|$; C11 is a steering claim proxied by null-removal AS; C10 uses class-text AS on 50 images/class (SING used ≈30 concepts / ≈1300 images). (ii) DenseNet has $\rho_{\mathrm{null}}{=}1.8{\times}10^{-8}$, so $\mathrm{AS}_{\mathrm{fair}}$ is degenerate. (iii) C13 (EVA02 as target space) is untestable: no EVA02-space translators were released. (iv) Our DinoViT IS/|AS| is 20.95 vs. 70.16 in Fig. 6b—absolute ratios differ, but the $\mathrm{AS}_{\mathrm{SING}}$ ranking still agrees.

Under $\mathrm{AS}_{\mathrm{fair}}$, DinoViT sits near the worst end (12th of 13 by class $\mathrm{AS}_{\mathrm{fair}}$; EVA02 worst); under $\mathrm{AS}_{\mathrm{SING}}$ it ranks 3rd. BiT becomes best under $\mathrm{AS}_{\mathrm{fair}}$ (0.042) because random subspaces of matched energy already move AS by 13.2°—the same translator-cancellation regime as Exp 1. DenseNet's near-zero ρ makes $\mathrm{AS}_{\mathrm{fair}}$ numerically unstable, so we also report τ without it (−0.364).

ID	Claim (short)	AS_{SING}	AS_{fair}
C1	DinoViT best IS/AS among 5	holds	reverses
C2	ResNeXt lowest IS/AS among 5	holds	reverses
C3	ResNet leaks more than DinoViT	holds	reverses
C4	DinoViT least class leakage	holds	reverses
C5	ResNeXt high AS (top-2 of 5)	holds	reverses
C6	ResNet per-class AS > DinoViT	holds	reverses
C7	DinoViT per-class |AS|<1	holds	reverses
C8	Porcupine/sports-car ≥90th pct.	reverses	reverses
C9	No ResNet–DinoViT class corr.	holds	reverses
C10	Jellyfish ≫ camel (DinoViT)	reverses	reverses
C11	DinoViT least steerable (proxy)	holds	reverses
C12	Fig. 6b 13-model order	holds	reverses
C13	EVA02 target-space order	untestable	untestable

Table 2: E2 claim audit. C1/C2/C11 use |AS| proxies; C10 is class-text on 50 imgs/class.

8 Discussion

SING's within-model AS is a useful probe of translator-mediated null content. Used across models, it conflates semantics with how large and energetic the null is. A boring baseline already said as much: over 13 models, Spearman($\rho_{\text{full}}, |\text{AS}|_{\text{full}}$)=0.87 and Spearman($\rho_{\text{full}}$, IS/AS)=−0.94. On top of that, the DinoViT label has no public checkpoint behind it.

Role of AS_{fair}. It divides null AS by an energy-matched random baseline so models are not rewarded for a small or low-energy null. E3 shows that neither score recovers planted α rankings at the pre-registered $\tau \geq 0.8$ bar, and AS_{fair} is worse at that particular task. We do not claim any replacement score beyond this size correction.

Practical suggestions. For any null-space leakage comparison:

1. Name the exact checkpoint and measured top-1 (avoid unverifiable labels such as "DinoViT").
2. Report d_{null} and ρ_{null} next to the score.
3. Report AS_{fair} (or another energy-matched baseline) beside raw AS.
4. Flag translator condition numbers when full-null and block-wise AS disagree (BiT).
5. Prefer within-model interventions (subspace size, planted signal) over cross-architecture league tables when d and ρ vary widely.

Limitations. On energy vs. dimension, E1 leaves the question partly open. E3 shows AS_{fair} does not recover planted α. Several E2 claims use proxies (C1/C2/C10/C11); DenseNet's AS_{fair} is degenerate; C13 is untestable. ImageNet-derived features restrict redistribution. We did not retrain translators, evaluate a true DINO ViT-B/16 [1], or extend the size audit to ViM [9] / Cook [2] / Rezaei [6] scores [4].

9 Conclusion

SING's AS ranking of 13 classifiers reproduces with the released translators, but it is size-confounded. Correcting for null energy with AS_{fair} breaks the published order (τ: 0.82 → −0.18) and reverses multiple claims, and the "DinoViT" result cannot be pinned on DINO. Until a calibrated leakage oracle exists, cross-model null-space comparisons should report ρ and a size-fair baseline alongside raw AS.

Reproducibility. Analysis code and aggregated tables: https://github.com/sudeshgm/ic014-sing-null-dimension. ImageNet features are private.

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A Additional numerical detail

Exp 1 full-null anchors. Mean |AS| at full null: ResNet50 1.682°, ResNeXt101 1.999°, BiT 0.556°, DINOv2-L 1.499°; at d=24: 0.166, 0.175, 9.30, 0.143 (seed SD in RESULTS.md).

E2 per-model class scores. Table 3 lists ρ_{null}, AS_{SING}, AS_{rand} and AS_{fair} used for the ranking.

Model	ρ_{null}	AS_{SING}	AS_{rand}	AS_{fair}
VGG16	0.350	1.088	1.537	0.708
VGG19	0.339	1.256	1.755	0.716
DenseNet121	1.8e-8	0.00025	0.0016	0.152
ResNet50	0.146	1.676	1.960	0.855
DinoViT	0.0054	0.294	0.180	1.632
EffNet-B0	0.130	0.443	0.498	0.889
BiT-R50x1	0.046	0.557	13.24	0.042
ResNeXt101	0.451	1.999	2.829	0.707
ConvNeXt-B	0.0135	0.080	0.083	0.974
Swin-L	0.274	1.019	1.229	0.829
DINOv2-L	0.322	1.499	2.105	0.712
DeiT3-L	0.348	5.048	4.738	1.065
EVA02-L	0.0089	0.417	0.160	2.612

Table 3: E2 class-text scores on the 10k subset (from e2_models.csv).

E3 per-concept τ. AS_{SING} / AS_{fair}: red 0.386 / 0.070; striped 0.403 / 0.061; water 0.369 / 0.083.